

TAN SIAO WAH

Motivated Artificial Intelligence graduate with research experience in Natural Language Processing and Computer Vision. Proficient in Python, PyTorch, TensorFlow, and Hugging Face Transformers, with strong skills in few-shot learning, hyperparameter tuning, and dataset processing. Published in IEEE Access and a conference, seeking an entry-level role in Artificial Intelligence to apply academic knowledge and contribute to impactful AI solutions.

Phone: +6011-6520 5895

Nationality: Malaysian

Email: 1207tansiaowah1207@gmail.com



WORK/PROJECT EXPERIENCE

Data Augmentation Enhanced Few-Shot Question Answering With Deep Learning

Master Research Project (2023-2025)

- Conducted research on few-shot extractive question answering (QA) to address performance limitations under low-resource training settings.
- Fine-tuned transformer-based models including T5, BART and Longformer to generate accurate answers from given questions and contexts.
- Designed a two-stage framework for QA, applying rule-based augmentation in pretraining and generative augmentation in fine-tuning.
- Performed systematic hyperparameter tuning to optimise model performance.
- Incorporated the trained models into an interactive QA system using Django, enabling users to input question and context (via text or Wikipedia title) and receive accurate context-based answer.

Food Detection And Recognition Using Machine Learning Final Year Project (2021-2022)

- Conducted a comparative study of deep learning models for multi-class food detection, including YOLOv5 and Faster R-CNN.
- Built end-to-end training pipelines, including dataset preprocessing, annotation conversion, hyperparameter tuning, image augmentation, and model evaluation using mean Average Precision (mAP) metric.

Artificial Intelligence Engineer Intern SmartPeep Sdn Bhd, Johor (Mar 2021 – Jun 2021)

- Contributed to the Homefall system by integrating an object detection model with an environmental mapping approach in Golang, reducing false detections in human fall detection.
- Developed and trained human detection models with TensorFlow, including data collection, data preprocessing, data augmentation, model training and evaluation.
- Improved detection reliability by analysing failure cases and refining the model.

EDUCATION

Master of Science (Information Technology) (By Research)

Multimedia University, Melaka (2023–2025)

Bachelor of Information Technology (Hons.) Artificial Intelligence

Multimedia University, Melaka (2019–2022)

- CGPA: 3.94 (First Class Honours)

PUBLICATIONS

- QARR-QGF: A Dual-Module Data Augmentation Framework for Enhanced Few-Shot Question Answering
- QARR-FSQA: Question-Answer Replacement and Removal Pretraining Framework for Few-Shot Question Answering
- Food Detection and Recognition with Deep Learning: A Comparative Study

SKILLS

- Programming & Tools: Python, Hugging Face, OpenCV, Git, C++, HTML, CSS, Docker, n8n.
- Machine Learning & AI: Natural Language Processing (NLP), Computer Vision (CV), PyTorch, TensorFlow, Few-Shot Learning, Object Detection, Data Preprocessing, Ground Truth Annotation, Data Augmentation, Model Fine-Tuning, Hyperparameter Tuning, Model Evaluation.
- Languages: Mandarin, English, Bahasa Malaysia (speaking, reading and writing).